Machine Learning with Big Data

Assignment 3

Recommender Systems using Content-Based and Collaborative Filtering on the MovieLens Dataset

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GitHub Link – Assignment 3

Introduction

The MovieLens dataset is a widely used benchmark dataset for recommender system research. It contains user ratings for movies along with movie metadata such as title and genres.

For this assignment, the **MovieLens Latest Small dataset** was used. The dataset contains approximately 100,000 ratings from around 600 users on nearly 9,000 movies.

The following files were used:

movies.csv – contains movieId, movie title, and genres
ratings.csv – contains userId, movieId, rating, and timestamp

Genres were treated as textual descriptions of movies and used as input features for building content-based recommendation models.

Step 1: Creating the folder structure

Name	Status	Date modified	Type	Size
dataset		17-03-2026 21:41	File folder	
Assignmnet3_Recommender		17-03-2026 21:47	Jupyter Source File	0 KB
M25DE1028_CSL7110_Assignment3		17-03-2026 21:57	Microsoft Word D...	350 KB
README		17-03-2026 21:43	Markdown Source ...	0 KB
requirements		17-03-2026 21:41	Text Document	0 KB

Step 2: Initialize git and first commit

```
aksha@AkshayKumar MINGW64 ~/OneDrive/Desktop/IITJ/Machine Learning with Big Data/Assignmnet 3/CSL7110_Assignment3_RecommenderSystems (master)
$ git status
fatal: not a git repository (or any of the parent directories): .git

aksha@AkshayKumar MINGW64 ~/OneDrive/Desktop/IITJ/Machine Learning with Big Data/Assignmnet 3/CSL7110_Assignment3_RecommenderSystems
$ git init
Initialized empty Git repository in C:/Users/aksha/OneDrive/Desktop/IITJ/Machine Learning with Big Data/Assignmnet 3/CSL7110_Assignment3_Recommende
Systems/.git/

aksha@AkshayKumar MINGW64 ~/OneDrive/Desktop/IITJ/Machine Learning with Big Data/Assignmnet 3/CSL7110_Assignment3_RecommenderSystems (master)
$ git add .

aksha@AkshayKumar MINGW64 ~/OneDrive/Desktop/IITJ/Machine Learning with Big Data/Assignmnet 3/CSL7110_Assignment3_RecommenderSystems (master)
$ git commit -m "Initial project structure for Assignment 3"
[master (root-commit) 09b35b6] Initial project structure for Assignment 3
 4 files changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 Assignmnet3_Recommender.ipynb
 create mode 100644 M25DE1028_CSL7110_Assignment3.docx
 create mode 100644 README.md
 create mode 100644 requirements.txt
```

Step 3: Connect GitHub

```
aksha@AkshayKumar MINGW64 ~/OneDrive/Desktop/IITJ/Machine Learning with Big Data/Assignmnet 3/CSL7110_Assignment3_RecommenderSystems (master)
$ git branch -M main

aksha@AkshayKumar MINGW64 ~/OneDrive/Desktop/IITJ/Machine Learning with Big Data/Assignmnet 3/CSL7110_Assignment3_RecommenderSystems (main)
$ git remote add origin https://github.com/Akshaykumarky26/CSL7110_Assignment3_RecommenderSystems.git

aksha@AkshayKumar MINGW64 ~/OneDrive/Desktop/IITJ/Machine Learning with Big Data/Assignmnet 3/CSL7110_Assignment3_RecommenderSystems (main)
$ git push -u origin main
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 24 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (6/6), 246.09 KiB | 12.30 MiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Akshaykumarky26/CSL7110_Assignment3_RecommenderSystems.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
```

Step 4: Download Dataset and push

```
aksha@AkshayKumar MINGW64 ~/OneDrive/Desktop/IITJ/Machine Learning with Big Data/Assignmnet 3/CSL7110_Assignment3_RecommenderSystems (main)
$ git add dataset

aksha@AkshayKumar MINGW64 ~/OneDrive/Desktop/IITJ/Machine Learning with Big Data/Assignmnet 3/CSL7110_Assignment3_RecommenderSystems (main)
$ git commit -m "Added MovieLens dataset"
[main d8dadf1] Added MovieLens dataset
 4 files changed, 124007 insertions(+)
 create mode 100644 dataset/links.csv
 create mode 100644 dataset/movies.csv
 create mode 100644 dataset/ratings.csv
 create mode 100644 dataset/tags.csv

aksha@AkshayKumar MINGW64 ~/OneDrive/Desktop/IITJ/Machine Learning with Big Data/Assignmnet 3/CSL7110_Assignment3_RecommenderSystems (main)
$ git push
Enumerating objects: 8, done.
Counting objects: 100% (8/8), done.
Delta compression using up to 24 threads
Compressing objects: 100% (7/7), done.
Writing objects: 100% (7/7), 950.91 KiB | 7.10 MiB/s, done.
Total 7 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/Akshaykumarky26/CSL7110_Assignment3_RecommenderSystems.git
   01b35c3..d8dadf1  main -> main
```

TASK 1: TF-IDF Based Content Recommender

In this task, a content-based recommender system was implemented using TF-IDF (Term Frequency–Inverse Document Frequency) on movie genres. The genres associated with each movie were treated as textual features and transformed into numerical vectors using TF-IDF vectorization.

The similarity between movies was computed using cosine similarity, which measures the angle between two vectors. A higher cosine similarity indicates greater similarity between movies.

A recommendation function was developed that takes a movie title as input and returns the top-N most similar movies along with their similarity scores.

The model successfully captures genre-based similarity and provides meaningful recommendations for movies sharing similar themes and categories.

```python
recommend_movies("Toy Story (1995)")
```

Python

```
(                                          title
 1706                                   Antz (1998)
 2355                             Toy Story 2 (1999)
 2809  Adventures of Rocky and Bullwinkle, The (2000)
 3000                Emperor's New Groove, The (2000)
 3568                          Monsters, Inc. (2001),
 [(1706, 1.0), (2355, 1.0), (2809, 1.0), (3000, 1.0), (3568, 1.0)])
```

TASK 2: User Profile Based Recommender

In this task, a personalized content-based recommender system was developed by constructing user profiles from historical ratings. Each user profile was computed as a weighted average of TF-IDF vectors of movies rated by the user.

The weights correspond to the ratings given by the user, ensuring that highly rated movies contribute more to the profile. The user profile vector captures the overall preferences of the user in terms of genres.

Cosine similarity was then computed between the user profile and all movie vectors to generate recommendations. Movies were ranked based on similarity scores and the top-N recommendations were returned.

This approach introduces personalization into the recommendation process and performs better than basic content-based filtering.

```python
recommend_for_user(1)
```

✓ 0.0s Python

	title
8597	Dragonheart 2: A New Beginning (2000)
6570	Hunting Party, The (2007)
4005	Flashback (1990)
4681	The Great Train Robbery (1978)
4409	Charlie's Angels: Full Throttle (2003)

TASK 3: User-Based Collaborative Filtering

In this task, a user-based collaborative filtering approach was implemented using the Surprise library. The model computes similarity between users based on their rating patterns.

A K-Nearest Neighbours (KNN) algorithm with cosine similarity was used to identify similar users. The predicted rating for a user-item pair was computed as a weighted average of ratings from similar users.

The model was evaluated using Root Mean Square Error (RMSE), which measures the difference between predicted and actual ratings.

Based on predicted ratings, top-N recommendations were generated for each user. This approach captures collaborative behaviour and provides more personalized recommendations compared to content-based methods.

```
recommend_cf(1)
✓ 0.0s                                                    Python
['The Jinx: The Life and Deaths of Robert Durst (2015)',
 'Galaxy of Terror (Quest) (1981)',
 'Alien Contamination (1980)',
 'Elling (2001)',
 'Sherlock - A Study in Pink (2010)']
```

TASK 4: Model-Based Collaborative Filtering (SVD)

In this task, a model-based collaborative filtering approach was implemented using Singular Value Decomposition (SVD). Unlike memory-based methods, SVD learns latent representations for users and items.

The model decomposes the user-item interaction matrix into lower-dimensional latent factors, capturing hidden relationships between users and movies. These latent factors represent implicit preferences and characteristics.

The predicted rating is computed using a combination of global average rating, user bias, item bias, and interaction between latent vectors.

The model was evaluated using RMSE and demonstrated improved accuracy compared to memory-based methods. This approach effectively handles sparsity and provides high-quality personalized recommendations.

```
recommend_svd(1)
✓ 0.0s                                                    Python
['Shawshank Redemption, The (1994)',
 'Dark Knight, The (2008)',
 'Rear Window (1954)',
 'North by Northwest (1959)',
 'Casablanca (1942)']
```

Conclusion

In this assignment, different recommender system techniques were implemented and compared using the MovieLens dataset. First, content-based filtering was performed using TF-IDF on movie genres, which helped in finding movies with similar characteristics. Then, a user-profile-based recommender was created to make the recommendations more personalized based on the movies rated by a user. After that, collaborative filtering methods were applied, where recommendations were generated using similarities between users and with the help of SVD, which is a model-based approach.

From the results, it was observed that each method has its own advantages. Content-based filtering works well when item features are available, while collaborative filtering is better for capturing patterns from user ratings. SVD gave better performance in terms of prediction accuracy and produced more personalized recommendations. Overall, this assignment helped in understanding how recommender systems work, how different techniques are implemented, and how they can be compared using evaluation metrics like RMSE.

Main learnings from this assignment:

- Understood the concept of content-based and collaborative filtering
- Learned how TF-IDF and cosine similarity can be used for recommendations
- Built user-profile-based personalized recommendations
- Implemented user-based collaborative filtering and SVD
- Compared different recommender models using RMSE
- Gained practical understanding of recommendation system design